



► LA RANCE BARRAGE
Opened in 1966, La Rance Barrage in St. Malo was the world's first tidal power station, and has been running reliably ever since.

WOW!

The rotating blades of an offshore wind turbine span up to 330 ft (100 m)—the length of seven school buses.



TIDAL FLOW

Moving water is incredibly powerful, but it is difficult to harness at sea. The most effective way is to use the power of the tides as they flow in and out of a river estuary. At St. Malo in France, the rising tide is allowed through a dam across the river mouth, and when the tide falls again, the water pouring out through the dam generates electricity.

OCEAN CURRENTS

Currents in oceans are like giant rivers flowing across the globe. In the future, it may be possible to use a powerful current such as the Gulf Stream to drive submerged rotors linked to electricity generators. Such a system could provide as much electricity as a nuclear power plant.

► RAISED ROTORS

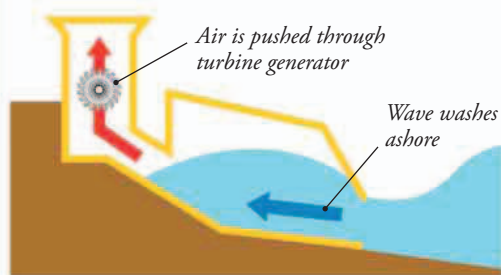
This pair of current turbines has been raised above sea level for maintenance.



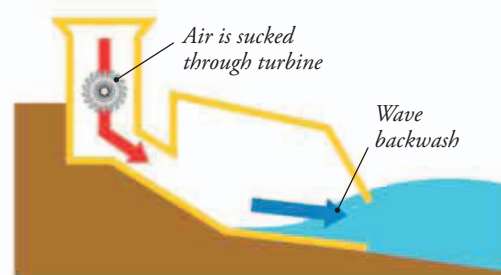
WAVE ENERGY

Ocean waves are very powerful, but can be very destructive. Turning that power into useful energy is not easy. The most successful systems use waves to pump air through pipes, creating high pressure

that drives turbine generators. These generators produce electricity in both directions—when the wave washes into the system, and when it washes back out again.



Inflow phase



Outflow phase

